‡‡‡ Chapter 31 was amended by **Local Law 77 of 2023**. This law has an effective date of June 10, 2023

CHAPTER 31

SPECIAL CONSTRUCTION

SECTION BC 3101
GENERAL

3101.1 Scope. The provisions of this chapter shall govern special building construction including membrane structures, temporary structures, pedestrian walkways and tunnels, automatic vehicular gates, awnings, canopies, sun control devices, marquees, signs, telecommunications towers and antennas, swimming pools and enclosures, sidewalk cafés, and fences.

SECTION BC 3102
MEMBRANE STRUCTURES

3102.1 General. The provisions of Sections 3102.1 through 3102.8 shall apply to air-supported structures, air-inflated structures, membrane-covered cable structures, membrane-covered frame structures and tents, and tensile membrane structures, collectively known as membrane structures.

3102.1.1 Certificates of Occupancy. The duration of Certificates of Occupancy for air-inflated structures, air-supported structures, and tents shall be limited in accordance with the requirements of Chapter 1 of Title 28 of the *Administrative Code*.

3102.1.2 Temporary installations. In addition to the requirements of Section 3102 of this code, temporary installations of air-supported structures, air-inflated structures, membrane-covered cable structures, tensile membrane structures, membrane-covered frame structures and tents, collectively known as membrane structures shall comply with the requirements of Article 111 of Chapter 1 of Title 28 of the *Administrative Code*.

3102.2 Definitions. The following terms are defined in Chapter 2:

AIR-INFLATED STRUCTURE.

AIR-SUPPORTED STRUCTURE.

Double skin.

Single skin.

CABLE-RESTRAINED, AIR-SUPPORTED STRUCTURE.

MEMBRANE-COVERED CABLE STRUCTURE.

MEMBRANE-COVERED FRAME STRUCTURE.

NONCOMBUSTIBLE MEMBRANE STRUCTURE.

TENSILE MEMBRANE STRUCTURE.

TENT.

3102.3 Type of construction. Noncombustible membrane structures shall be classified as Type IIB construction. Noncombustible frame or cable-supported structures covered by an approved membrane in accordance with Section 3102.3.1 shall be classified as Type IIB construction. Heavy timber frame-supported structures covered by an approved membrane in accordance with Section 3102.3.1 shall be classified as Type IV construction. Other membrane structures shall be classified as Type V construction.

3102.3.1 Membrane and interior liner material. Membranes and interior liners shall be either noncombustible as set forth in Section 703.5, or meet the fire propagation performance criteria of Test Method 1 or Test Method 2, as appropriate, of NFPA 701 and the manufacturer's test protocol.

3102.4 Allowable floor areas. The area of a membrane structure shall not exceed the limitations specified in Section 506.

3102.5 Maximum height. Membrane structures shall not exceed one story nor shall such structures exceed the height limitations in feet specified in Section 504.3. Membrane structures may be erected above the roof of a building provided that such roof is of noncombustible construction required to have a fire-resistance rating.

Exception: Noncombustible membrane structures serving as roofs only.

3102.6 Mixed construction. Membrane structures shall be permitted to be utilized as specified in this section as a portion of buildings of other types of construction. Height and area limits shall be as specified for the type of construction and occupancy of the building.

3102.6.1 Noncombustible membrane. A noncombustible membrane shall be permitted for use as the roof or as a skylight of any building or atrium of a building of any type of construction provided such membrane is not less than 20 feet (6096 mm) above any floor, balcony or gallery, and meets the fire classification requirements of Section 1505 for roof assemblies.

3102.6.1.1 Membrane. A membrane meeting the fire propagation performance criteria of Test Method 1 or Test Method 2, as appropriate, of NFPA 701 shall be permitted to be used as the roof or as a skylight on buildings of Type IIB, III, IV and V construction, provided such membrane is not less than 20 feet (6096 mm) above any floor, balcony or gallery, and meets the fire classification requirements of Section 1505 for roof assemblies.

3102.7 Engineering design. Membrane structures, except tensile membrane structures, shall be designed and constructed to sustain dead loads; loads due to tension or inflation; live loads including wind, snow or flood and seismic loads and in accordance with Chapter 16 and Appendix G.

3102.7.1 Lateral restraint. For membrane-covered frame structures, the membrane shall not be considered to provide lateral restraint in the calculation of the capacities of the frame members.

3102.7.2 Design. Tents, air-inflated structures, and air-supported structures shall be designed by either an alternate engineering design approved by the commissioner or as follows:

1. Tents. Tents shall be guyed, supported, and braced to withstand a wind pressure of 10 psf (478.8 Pa) of projected area of the tent. The poles and their supporting guys, stays, stakes, fastenings, etc., shall be of sufficient strength and attached so as to resist wind pressure of 20 psf (957.6 Pa) of projected area of the tent.
2. Air-inflated structures and air-supported structures.

   2.1 Air-inflated structures and air-supported structures shall be anchored to the ground or supporting structure by either ballast or positive anchorage, sufficiently and evenly distributed, and adequate to resist the inflation lift load, the aerodynamic lift load, and the drag (shear) load due to wind impact. The latter factors shall be based on a fastest mile wind speed of 70 mph (112.65 km/hr), and an estimated stagnation of not less than 0.5 q for structures on grade whose height is equal to, or less than, the width of the structure. For greater heights, or for elevated structures, increased anchorage shall be provided, justified by analytical and/or experimental data subject to approval by the commissioner.

   2.2 The skin of the structure shall be of such strength, and the joints so constructed, as to provide a minimum dead load strip tensile strength at 70°F (21°C) of four times the 70 mph (121.65 km/hr) design load (inflation and aerodynamic loading). The joints shall provide a dead load strip tensile strength of 160°F (71°C) of twice the 70 mph (121.65 km/hr) design load (i.e., a factor of safety of 4 and 2, respectively). In addition, the material shall provide a trapezoidal tear strength of not less than 15 percent of the maximum design tensile load. Material and joint strengths shall be so certified by the manufacturer, justified by analytical and/or experimental data.

3102.7.3 Tensile membrane structures. The design and construction shall be in accordance with ASCE 55. The provisions of Sections 3102.1 through 3102.6 shall also apply.

3102.8 Inflation systems. Air-supported structures and air-inflated structures shall be provided with primary and auxiliary inflation systems to meet the minimum requirements of Sections 3102.8.1 through 3102.8.3.

3102.8.1 Equipment requirements. The primary inflation system shall consist of one or more blowers and shall include provisions for automatic control to maintain the required inflation pressures. Such system shall be so designed as to prevent overpressurization of the system.

3102.8.1.1 Auxiliary inflation system. In addition to the primary inflation system, in structures exceeding 1,500 square feet (140 m^2) in area, an auxiliary inflation system shall be provided with sufficient capacity to maintain the inflation of the structure in case of primary system failure. The auxiliary inflation system shall operate automatically when there is a loss of internal pressure and when the primary blower system becomes inoperative. Initiation of auxiliary inflation system shall be indicated by an audible and visual alert at the entrance or a regularly attended location.

3102.8.1.2 Blower equipment. Blower equipment shall meet all of the following requirements:

1. Blowers shall be powered by continuous-rated motors at the maximum power required for any flow condition as required by the structural design.
2. Blowers shall be provided with inlet screens, belt guards and other protective devices as required by the commissioner to provide protection from injury.
3. Blowers shall be housed within a weather-protecting structure.
4. Blowers shall be equipped with backdraft check dampers to minimize air loss when inoperative.
5. Blower inlets shall be located to provide protection from air contamination. The location of inlets shall be approved.

3102.8.2 Standby power. Wherever an auxiliary inflation system is required, an approved standby power-generating system shall be provided. However, notwithstanding Section 2702.1, the standby power-generating system shall be equipped with a suitable means for automatically starting the generator set upon failure of the normal electrical service and for automatic transfer and operation of all of the required electrical functions at full power within 60 seconds of such service failure. Standby power shall be capable of operating independently for not less than 4 hours. Initiation of standby power shall be indicated by an audible and visual alert at the entrance or a regularly attended location.

3102.8.3 Support provisions. A system capable of supporting the membrane in the event of deflation shall be provided for in air-supported structures and air-inflated structures having an occupant load of 50 or more or where covering a swimming pool regardless of occupant load. Such support system shall be capable of maintaining the membranes not less than 7 feet (2134 mm) above the floor, seating area or surface of the water. When air-supported structures or air-inflated structures are used as a roof on Type I or II construction buildings, such support system shall be capable of maintaining the membranes not less than 20 feet (6096 mm) above the floor or seating area.

3102.9 Separation. No air-inflated structure, air-supported structure, or tent shall be erected closer than 20 feet (6096 mm) to any interior lot line nor closer than 30 feet (9144 mm) in any direction to an unprotected opening, required exterior stairway or corridor, or required exit door, on the same level or above the level of such structure. Such structure may abut another building on the same tax lot if the following conditions exist:

1. No unprotected openings or exits are located above or within 30 feet (9144 mm) of such structure.
2. No doors serving as a required exit are located between such structure and the abutted building.
3. The exterior wall of the abutted building meets the requirements of Section 705 for fire walls.

3102.10 Exits. In addition to the requirements of Chapter 10, travel distance to an exit from any point within a tent, air-supported structure, or air-inflated structure shall not exceed 75 feet (22 860 mm).

3102.10.1 Exit openings from tents. Exit openings from tents shall remain open unless covered by a flame-resistant curtain of a contrasting color to the tent. Such curtain shall be supported not less than 80 inches (2032 mm) above the floor level at the exit and, when open, no part of the curtain shall obstruct the exit.

3102.10.2 Exit openings from air-supported structures and air-inflated structures. Exit doors in air-supported structures and air-inflated structures shall close automatically against normal operational pressures. Opening force at the edge of such doors shall not exceed 15 pounds (6.80 kg), with the structure at operational pressure. Exit doors shall be located in frames constructed such that they will remain operative and support the weight of the pressurized membrane structure in a state of total collapse.

SECTION BC 3103
TEMPORARY STRUCTURES

3103.1 General. The provisions of Sections 3103.1 through 3103.4 shall apply to temporary tents, grandstands, platforms, reviewing stands, outdoor bandstands, stages and similar miscellaneous structures erected for a period of 90 days or less. Such structures may be constructed of wood whether located inside or outside of the fire districts.

3103.1.1 Permit required. Temporary structures shall not be erected, operated or maintained for any purpose without obtaining a permit from the department in accordance with Section 28-111.1.1 of the *Administrative Code*.

Exception: No permit shall be required for:

1. The erection and use of temporary tents of less than 400 gross square feet (37 m^2), for not more than 30 days.
2. The erection and use of temporary platforms, reviewing stands, outdoor bandstands, and similar miscellaneous structures that cover an area less than 120 square feet (11.16 m^2), including connecting areas or spaces with a common means of egress or entrance, for not more than 30 days.

3103.2 Construction documents. A permit application and construction documents shall be submitted for each installation of a temporary structure. The construction documents shall include a site plan indicating the location of the temporary structure and information delineating the means of egress and the occupant load.

3103.3 Location. Temporary structures shall be located in accordance with the requirements of Table 602 based on the fire-resistance rating of the exterior walls for the proposed type of construction.

3103.4 Means of egress. Temporary structures shall conform to the means of egress requirements of Chapter 10 and shall have an exit access travel distance of 100 feet (30 480 mm) or less.

SECTION BC 3104
PEDESTRIAN WALKWAYS AND TUNNELS

3104.1 General. This section shall apply to connections between buildings such as pedestrian walkways or tunnels, located at, above or below grade level, that are used as a means of travel by persons. The pedestrian walkway shall not contribute to the building area or the number of stories or height of connected buildings.

3104.1.1 Application. Pedestrian walkways shall be designed and constructed in accordance with Sections 3104.2 through 3104.9. Tunnels shall be designed and constructed in accordance with Sections 3104.2 and 3104.10.

3104.2 Separate structures. Buildings connected by pedestrian walkways or tunnels shall be considered to be separate structures.

Exceptions:

1. Buildings that are on the same tax lot and considered as portions of a single building in accordance with Section 503.1.2.
2. For purposes of calculating the number of Type B units required by Chapter 11, structurally connected buildings and buildings with multiple wings shall be considered one structure.

3104.3 Construction. The pedestrian walkway shall be of a Construction Type that is at least equal to the higher type of the two buildings connected.

Exception: Exterior pedestrian walkways serving as a required exit shall be constructed of noncombustible materials and comply with the fire-resistance rating requirements as set forth in Chapter 7.

3104.4 Contents. Only materials approved by the department shall be located in the pedestrian walkway. Decorations may be permitted in accordance with the *New York City Fire Code*.

3104.5 Connections of pedestrian walkways to buildings. The connection of a pedestrian walkway to a building shall comply with Section 3104.5.1, 3104.5.2, 3104.5.3 or 3104.5.4.

Exception: Buildings that are on the same tax lot and considered as portions of a single building in accordance with Section 503.1.2.

3104.5.1 Fire barriers. Pedestrian walkways shall be separated from the interior of the building by not less than 2-hour fire barriers constructed in accordance with Section 707 and Sections 3104.5.1.1 through 3104.5.1.3.

3104.5.1.1 Exterior walls. Exterior walls of buildings connected to pedestrian walkways shall be 2-hour fire-resistance-rated. This protection shall extend not less than 10 feet (3048 mm) in every direction surrounding the perimeter of the pedestrian walkway.

3104.5.1.2 Openings in exterior walls of connected buildings. Openings in exterior walls required to be fire-resistance-rated in accordance with Section 3104.5.1.1 shall be equipped with opening protectives providing a not less than $^{3}/_{4}$-hour fire protection rating in accordance with Section 716.

3104.5.1.3 Supporting construction. The fire barrier shall be supported by construction as required by Section 707.5.1.

3104.5.2 Alternative separation. The wall separating the pedestrian walkway and the building shall comply with Section 3104.5.2.1 or 3104.5.2.2 where:

1. The distance between the connected buildings is more than 30 feet (9144 mm).
2. The pedestrian walkway and connected buildings are equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1, and the roof of the walkway is not more than 55 feet (16 764 mm) above grade connecting to the fifth, or lower, story above grade plane, of each building.

Exception: Pedestrian walkways accessory to open parking garages need not be equipped with an automatic sprinkler system.

3104.5.2.1 Passage of smoke. The wall shall be capable of resisting the passage of smoke.

3104.5.2.2 Glass. The wall shall be constructed of a tempered, wired or laminated glass wall and doors or glass separating the interior of the building from the pedestrian walkway. The glass shall be protected by an automatic sprinkler system in accordance with Section 903.3.1.1 that, when actuated, shall completely wet the entire surface of interior sides of the wall or glass. Obstructions shall not be installed between the sprinkler heads and the wall or glass. The glass shall be in a gasketed frame and installed in such a manner that the framing system will deflect without breaking (loading) the glass before the sprinkler operates.

3104.5.3 Open sides on walkway. Where the distance between the connected buildings is more than 30 feet (9144 mm), the walls at the intersection of the pedestrian walkway and each building need not be fire-resistance-rated provided both sidewalls of the pedestrian walkway are not less than 50 percent open with the open area uniformly distributed to prevent the accumulation of smoke and toxic gases. The roof of the walkway shall be located not more than 40 feet (12 160 mm) above grade plane, and the walkway shall only be permitted to connect to the third or lower story of each building.

Exception: Where the pedestrian walkway is protected with a sprinkler system in accordance with Section 903.3.1.1, the roof of the walkway shall be located not more than 55 feet (16 764 mm) above grade plane and the walkway shall only be permitted to connect to the fifth or lower story of each building.

3104.5.4 Exterior walls greater than 2 hours. Where exterior walls of connected buildings are required by Section 705 to have a fire-resistance rating greater than 2 hours, the walls at the intersection of the pedestrian walkway and each building need not be fire-resistance-rated provided:

1. The pedestrian walkway is equipped throughout with an automatic sprinkler system installed in accordance with Section 903.3.1.1.
2. The roof of the walkway is not located more than 55 feet (16 764 mm) above grade plane and the walkway connects to the fifth, or lower, story above grade plane of each building.

3104.6 Public way. Pedestrian walkways over a public way shall comply with Chapter 32.

3104.7 Width. The unobstructed width of pedestrian walkways shall be not less than 36 inches (914 mm). The total width shall be not greater than 30 feet (9144 mm).

3104.8 Egress. Access shall be provided at all times to a pedestrian walkway that serves as a required exit. Doors satisfying the requirements of Chapter 10 shall enclose each end of such pedestrian walkway. The width of such pedestrian walkway shall be at least equal to the width of the doors opening onto such pedestrian walkway, but in no case less than 44 inches (1118 mm). The floor level at doors shall be the same as that of the connected building.

Exception: The floor level at doors of open pedestrian walkways shall be not less than $7^1/_2$ inches (191 mm) below the level of the door. Where the requirements of Chapter 11 are applicable, these differences in level shall be accommodated by means of ramps in compliance with the provisions of Chapter 11.

3104.9 Exit access travel. The length of exit access travel shall be 200 feet (60 960 mm) or less.

Exceptions:

1. Exit access travel distance on a pedestrian walkway equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1 shall be 250 feet (76 200 mm) or less.
2. Exit access travel distance on a pedestrian walkway constructed with both sides not less than 50 percent open shall be 300 feet (91 440 mm) or less.
3. Exit access travel distance on a pedestrian walkway constructed with both sides not less than 50 percent open, and equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1, shall be 400 feet (122 m) or less.

3104.10 Tunneled walkway. Separation between the tunneled walkway and the building to which it is connected shall be not less than 2-hour fire-resistant construction and openings therein shall be protected in accordance with Table 716.5.

SECTION BC 3105
AWNINGS, CANOPIES
AND SUN CONTROL DEVICES

3105.1 General. Awnings, canopies, and sun control devices shall comply with the requirements of this section, the requirements of Chapter 32 for projections over public ways, and other applicable sections of this code.

Exception: Canopies projecting over public rights-of-way governed by Title 19 of the *Administrative Code* and rules of the *New York City Department of Transportation*.

3105.2 Definitions. The following term is defined in Chapter 2:

RETRACTABLE AWNING.

SUN CONTROL DEVICE.

3105.3 Design and construction. Awnings, canopies, and sun control devices shall be designed and constructed to withstand wind or other lateral loads and live loads as required by Chapter 16 with due allowance for shape, open construction and similar features that relieve the pressures or loads. Structural members shall be protected to prevent deterioration. Awnings and canopies shall have frames of noncombustible material, covered with fabric that meets the fire propagation performance criteria of Test Method 1 or Test Method 2, as appropriate, of NFPA 701 or has a flame spread index not greater than 25 when tested in accordance with ASTM E 84 or UL 723, or covered in plastic in accordance with Section 2605, sheet metal, or other equivalent material, and shall be either fixed, retractable, folding or collapsible. Sun control devices shall be constructed of noncombustible materials.

3105.4 Reserved.

SECTION BC 3106
MARQUEES

3106.1 General. Marquees shall comply with this section and other applicable sections of this code. Marquees projecting beyond the street line shall also comply with the requirements of Chapter 32. Signs placed on marquees shall also comply with Section 3107.

3106.2 Thickness. The maximum height or thickness of a marquee measured vertically from its lowest to its highest point shall not be limited.

Exception: Marquees projecting beyond the street line shall meet the height and thickness requirements of Chapter 32.

3106.3 Roof construction. Where the roof or any part thereof is a skylight, the skylight shall comply with the requirements of Chapter 24. Every roof and skylight of a marquee shall be drained in accordance with the provisions of the *New York City Plumbing Code*.

3106.4 Location prohibited. Every marquee shall be so located as not to interfere with the operation of any exterior standpipe, and such that the marquee does not obstruct the clear passage of stairways or exit discharge from the building or the installation or maintenance of street lighting.

3106.5 Construction. A marquee shall be supported entirely from the building and constructed of noncombustible materials. Marquees shall be designed as required in Chapter 16. Structural members shall be protected to prevent deterioration.

SECTION BC 3107
SIGNS

3107.1 General. Signs shall be designed, constructed and maintained in accordance with Appendix H.

SECTION BC 3108
RADIO, TELEVISION, AND TELECOMMUNICATIONS TOWERS AND ANTENNAS

3108.1 General. Subject to the provisions of Chapter 16 and the requirements of Chapter 15 governing the fire-resistance ratings of buildings for the support of roof structures, radio, television, and telecommunications towers and antennas shall be designed and constructed as herein provided. All such towers and antennas shall be collectively referred to as "towers" for the purposes of this section. Towers shall be designed and constructed in accordance with the provisions of TIA-222.

Exceptions:

1. Exceptions related to seismic design in Section 2.7.3 of TIA-222 shall not apply;
2. The specified values for seismic coefficients S_s, S_1 and Site Factors F_a and F_v shall comply with Section 1613.3; and
3. The horizontal extent of Topographic Category 2, escarpments, shall be 16 times the height of the escarpment.

3108.2 Location and access. Towers shall be located and equipped with step bolts and ladders so as to provide ready access for inspection purposes. Guy wires or other accessories shall not cross or encroach upon any street or other public space, or over above-ground electric utility lines, or encroach upon any privately owned property without written consent of the owner of the encroached-upon property, space or above-ground electric utility lines. Towers shall be equipped with climbing and working facilities in compliance with TIA-222. See applicable OSHA, FCC and EPA regulations relating to limitations on access to tower sites.

3108.3 Construction. Towers shall be constructed of approved corrosion-resistant noncombustible material. The minimum type of construction of isolated radio towers not more than 100 feet (30 480 mm) in height shall be Type IIB.

3108.4 Loads. Towers shall be designed to resist wind loads in accordance with TIA-222 and Chapter 16 of this code. The design criteria shall not be less stringent than required by Chapter 16. Consideration shall be given to conditions involving wind load on ice-covered sections.

3108.4.1 Dead load. Towers shall be designed for the dead load plus ice load.

3108.4.2 Wind load. Towers shall be provided with adequate foundations and anchorage designed to resist two times the calculated wind load.

3108.4.3 Earthquake load. Towers shall be provided with adequate foundations and anchorage designed to resist two times the calculated earthquake load.

3108.5 Grounding. Towers shall be permanently and effectively grounded in accordance with the *New York City Electrical Code*.

SECTION BC 3109
SWIMMING POOLS, SWIMMING POOL ENCLOSURES, AND SAFETY DEVICES

3109.1 General. Swimming pools, swimming pool enclosures, and swimming pool safety devices shall comply with the requirements of this section and other applicable sections of this code.

3109.2 Definitions. The following terms are defined in Chapter 2:

BARRIER, TEMPORARY.

SWIMMING POOL.

SWIMMING POOL, PRIVATE.

SWIMMING POOL, PUBLIC.

3109.3 Pool safety and accessibility. Public and private swimming pools shall comply with the requirements for safety and accessibility as provided in this section.

3109.3.1 Entrapment protection. Entrapment protection shall be provided in compliance with this section.

3109.3.1.1 Suction entrapment avoidance. Suction outlets shall be designed and installed in accordance with ANSI/APSP-7.

3109.3.2 Swimming pool and spa alarms. Swimming pool and spa alarms shall comply with Sections 3109.3.2.1 through 3109.3.2.4.

3109.3.2.1 Applicability. All swimming pools and spas shall be equipped with an approved pool alarm. Pool alarms shall comply with ASTM F 2208 and shall be installed, used and maintained in accordance with the manufacturer's instructions and this section.

Exceptions:

1. A hot tub or spa equipped with a safety cover that complies with ASTM F 1346.
2. A swimming pool (other than a hot tub or spa) equipped with an automatic power safety cover that complies with ASTM F 1346.

3109.3.2.2 Multiple alarms. A pool alarm must be capable of detecting entry into the water at any point on the surface of the swimming pool. If necessary to provide detection capability at every point on the surface of the swimming pool, more than one pool alarm shall be provided.

3109.3.2.3 Alarm activation. Pool alarms shall activate upon detecting entry into the water and shall sound poolside and inside the building.

3109.3.2.4 Prohibited alarms. The use of personal immersion alarms shall not be construed as compliance with this section.

3109.3.3 Water circulation, water treatment and drainage. The supply, circulation, treatment, and drainage of water for swimming pools shall meet the requirements of the *New York City Plumbing Code*.

3109.3.4 Electrical precautions. No overhead electrical conductors shall be installed within 15 feet (4572 mm) of any swimming pool. All metal fences, enclosures, or railings that might become electrically charged as a result of contact with broken overhead conductors or from any other cause near, or adjacent to, a swimming pool shall be grounded in accordance with the provisions of lightning protection in the *New York City Electrical Code*.

3109.3.5 Facilities for people with disabilities. Facilities for people with physical disabilities shall be provided where required by Chapter 11 of this code.

3109.4 Public swimming pools. Public swimming pools shall comply with the requirements for safety and accessibility as provided in Sections 3109.3 and 3109.4.

Exception: A swimming pool with a power safety cover or a spa with a safety cover complying with ASTM F 1346.

3109.4.1 Barrier height and clearances. Public swimming pools shall be completely enclosed by a fence, wall, building, or other solid barrier, or any combination thereof, not less than 6 feet (1829 mm) in height. Openings in the enclosure and pedestrian access gates shall not permit the passage of a 4-inch-diameter (102 mm) sphere. The enclosure shall be equipped with self-closing and self-latching gates.

Exception: Enclosures shall be not less than 4 feet (1219 mm) in height when surrounding wading pools with water less than 24 inches (610 mm) in depth.

3109.4.2 Gates. Gates shall comply with Sections 3109.4.2.1 through 3109.4.2.3.

3109.4.2.1 Self-closing; opening configuration. All gates shall be self-closing. In addition, if the gate is a pedestrian access gate, the gate shall open outward, away from the pool.

3109.4.2.2 Self-latching; location of latch handle. All gates shall be self-latching, with the latch handle located within the enclosure (i.e., on the pool side of the enclosure) and not less than 40 inches (1016 mm) above grade. If the latch handle is located less than 54 inches (1372 mm) from the bottom of the gate, the latch handle shall be located not less than 3 inches (76 mm) below the top of the gate, and neither the gate nor barrier shall have any opening greater than 0.5 inch (12.7 mm) within 18 inches (457 mm) of the latch handle.

3109.4.2.3 Locking. All gates shall be securely locked with a key, combination or other child proof lock sufficient to prevent access to the swimming pool through such gate when the swimming pool is not in use or supervised.

3109.4.3 Other laws. In addition to the requirements of this section, any other, more stringent requirements for the construction and design of swimming pool and barriers that may be provided for in Article 165 of the *New York City Health Code*, as administered by the New York City Department of Health and Mental Hygiene, shall also be applicable.

‡‡‡ **3109.5 Private swimming pools.** Private swimming pools shall comply with the requirements for safety and accessibility as provided in Section 3109.3 and this section.

Exception: An above-ground private swimming pool which has a maximum water depth of 4 feet (1219 mm) and an area not exceeding 500 square feet (46.45 m^2) that is accessory to an R-3 occupancy and is privately used for noncommercial purposes shall not be required to comply with Sections 3109.3.1, 3109.3.2, 3109.5.2, 3109.5.3 and 3109.5.4.

3109.5.1 Barrier height and clearances. The top of the barrier enclosing a private swimming pool shall be not less than 48 inches (1219 mm) above grade measured on the side of the barrier that faces away from the swimming pool. The vertical clearance between grade and the bottom of the barrier shall be not greater than 2 inches (51 mm) measured on the side of the barrier that faces away from the swimming pool. Where the top of the pool structure is above grade, the barrier is authorized to be erected at grade level or mounted on top of the pool structure, and the vertical clearance between the top of the pool structure and the bottom of the barrier shall be not greater than 4 inches (102 mm).

3109.5.1.1 Openings. Openings in the barrier shall not allow passage of a 4-inch-diameter (102 mm) sphere.

3109.5.1.2 Solid barrier surfaces. Solid barriers which do not have openings shall not contain indentations or protrusions except for normal construction tolerances and tooled masonry joints.

3109.5.1.3 Closely spaced horizontal members. Where the barrier is composed of horizontal and vertical members and the distance between the tops of the horizontal members is less than 45 inches (1143 mm), the horizontal members shall be located on the swimming pool side of the fence. Spacing between vertical members shall be not greater than $1^{3}/_{4}$ inches (44 mm) in width. Where there are decorative cutouts within vertical members, spacing within the cutouts shall be not greater than $1^{3}/_{4}$ inches (44 mm) in width.

3109.5.1.4 Widely spaced horizontal members. Where the barrier is composed of horizontal and vertical members and the distance between the tops of the horizontal members is 45 inches (1143 mm) or more, spacing between vertical members shall be not greater than 4 inches (102 mm). Where there are decorative cutouts within vertical members, spacing within the cutouts shall be not greater than $1^{3}/_{4}$ inches (44 mm) in width.

3109.5.1.5 Chain link dimensions. Mesh size for chain link fences shall be not greater than a $2^{1}/_{4}$ inch square (57 mm square) unless the fence is provided with slats fastened at the top or the bottom that reduce the openings to not more than $1^{3}/_{4}$ inches (44 mm).

3109.5.1.6 Diagonal members. Where the barrier is composed of diagonal members, the opening formed by the diagonal members shall be not greater than $1^{3}/_{4}$ inches (44 mm).

3109.5.1.7 Gates. Gates shall comply with Sections 3109.5.1.1 through 3109.5.1.6 and Sections 3109.5.1.7.1 through 3109.5.1.7.3.

3109.5.1.7.1 Self-closing; opening configuration. All gates shall be self-closing. In addition, if the gate is a pedestrian access gate, the gate shall open outward, away from the pool.

3109.5.1.7.2 Self-latching; location of latch handle. All gates shall be self-latching, with the latch handle located within the enclosure (i.e., on the pool side of the enclosure) and not less than 40 inches (1016 mm) above grade. In addition, if the latch handle is located less than 54 inches (1372 mm) from the bottom of the gate, the latch handle shall be located not less than 3 inches (76 mm) below the top of the gate, and neither the gate nor barrier shall have any opening greater than 0.5 inch (12.7 mm) within 18 inches (457 mm) of the latch handle.

3109.5.1.7.3 Locking. All gates shall be securely locked with a key, combination or other child proof lock sufficient to prevent access to the swimming pool through such gate when the swimming pool is not in use or supervised.

3109.5.1.8 Dwelling wall as a barrier. Where a wall of a dwelling serves as part of the barrier, one of the following shall apply:

1. Doors with direct access to the pool through that wall shall be equipped with an alarm that produces an audible warning when the door and/or its screen, if present are opened. The alarm shall be listed in accordance with UL 2017. The audible alarm shall activate within 7 seconds and sound continuously for a minimum of 30 seconds immediately after the door and/or its screen, if present, are opened and be capable of being heard throughout the house during normal household activities. The alarm shall automatically reset under all conditions. The alarm shall be equipped with a manual means, such as touchpad or switch, to temporarily deactivate the alarm for a single opening. Such deactivation shall last no more than 15 seconds. In dwellings not required to be accessible Type B units, the deactivation switch shall be located 54 inches (1372 mm) or more above the threshold of the door. In dwellings required to be accessible Type B units, the deactivation switch(es) shall be located at 48 inches (1219 mm) above the threshold of the door.
2. The swimming pool shall be equipped with a power safety cover which complies with ASTM F 1346.
3. The door providing access to the swimming pool from the dwelling shall open inward, away from the swimming pool, and shall be self-closing and have a self-latching device. The release mechanism of the self-latching device shall be located no less than 54 inches (1372 mm) from the bottom of the door. In dwellings required to be accessible Type B units, the release mechanism shall be located at 48 inches (1219 mm) above the threshold of the door.

3109.5.1.9 Pool structure as barrier. Where an above-ground private swimming pool structure is used as a barrier or where the barrier is mounted on top of the pool structure, and the means of access is a ladder or steps, then the ladder or steps either shall be capable of being secured, locked or removed to prevent access, or the ladder or steps shall be surrounded by a barrier that meets the requirements of Sections 3109.5.1.1 through 3109.5.1.8. Where the ladder or steps are secured, locked or removed, any opening created shall not allow the passage of a 4-inch-diameter (102 mm) sphere.

3109.5.2 Indoor swimming pools. Walls surrounding indoor private swimming pools shall not be required to comply with Section 3109.5.1.8.

3109.5.3 Prohibited locations. Barriers shall be located so as to prohibit permanent structures, equipment or similar objects from being used to climb the barriers.

3109.5.4 Construction requirements. Private swimming pools shall be constructed so as to be water tight and easily cleaned. They shall be built of nonabsorbent materials with smooth surfaces and shall be free of open cracks and open joints.

3109.5.4.1 Walls. The walls of swimming pools shall be vertical for not less than the top 30 inches (762 mm) below the normal water level. The junctions between the side walls and the bottom shall be coved. A swimming pool overflow shall be provided meeting the requirements of the *New York City Plumbing Code*.

3109.5.4.2 Bottom slopes. The bottom of the swimming pool shall slope downward toward the main drains. The slope in shallow areas with depths less than 5 feet (1524 mm) shall not exceed 1 unit vertical in 12 units horizontal (8-percent slope). In portions of the swimming pool with depth greater than 5 feet (1524 mm), the slope shall not be steeper than 1 unit vertical in 3 units horizontal (33-percent slope).

3109.5.4.3 Ladders. There shall be a ladder or steps with handrails at the deep end and at the shallow end of every swimming pool. Ladders and steps shall have no-slip treads. All ladders shall be rigidly installed and shall be constructed of corrosion-resistant materials.

3109.5.4.4 Walkways. Every swimming pool shall have a walkway not less than 5 feet (1524 mm) wide around its entire perimeter. The walkway shall have a nonslip surface and shall be constructed to drain away from the swimming pool.

3109.5.4.5 Handholds. Every swimming pool shall be constructed so that either the overflow gutter, if provided, or the tops of the side walls afford a continuous handhold for bathers.

3109.5.4.6 Markings. Permanent markings showing the depth of the shallow end, break points, diving depth and deep end shall be provided so as to be visible from both inside and outside the swimming pool.

3109.5.4.7 Diving boards and towers. Diving towers shall be rigidly constructed and permanently anchored. The depth of the water below a diving board shall be not less than 102 inches (2591 mm) for a board 39 inches (991 mm) or less above the water. For a diving board more than 39 inches (991 mm) and not more than 118 inches (2997 mm) above the water, the depth of the water below the board shall be not less than 144 inches (3658 mm). For a diving board or platform more than 118 inches (2997 mm) above the water, the depth of the water below the board shall be not less than 192 inches (4877 mm). Indoor swimming pools shall provide not less than 144 inches (3658 mm) overhead clearance above all diving boards.

3109.6 Temporary barriers. An outdoor swimming pool, including an in-ground, above ground or on-ground pool, hot tub or spa shall be surrounded by a temporary barrier during installation or construction. Such barrier shall remain in place until a permanent fence in compliance with Section 3109.4 is provided for public swimming pools, or a barrier in compliance with Section 3109.5 is provided for residential private pools.

Exceptions:

1. Above-ground or on-ground residential swimming pools where the pool structure is the barrier in compliance with Section 3109.
2. Spas or hot tubs with a safety cover which complies with ASTM F 1346, provided that such safety cover is in place during the period of installation or construction of such hot tub or spa. The temporary removal of a safety cover as required to facilitate the installation or construction of a hot tub or spa during periods when at least one person engaged in the installation or construction is present is permitted.

3109.6.1 Height. The top of the temporary barrier shall be not less than 48 inches (1219 mm) above grade measured on the side of the barrier which faces away from the swimming pool.

3109.6.2 Replacement by a permanent barrier. A temporary barrier shall be replaced by a complying permanent barrier within either of the following periods:

1. 90 days of the date of issuance of the building permit for the installation or construction of the swimming pool; or
2. 90 days of the date of commencement of the installation or construction of the swimming pool.

3109.6.2.1 Replacement extension. Subject to the approval of the code enforcement official, the time period for completion of the permanent barrier may be extended for good cause, including, but not limited to, adverse weather conditions delaying construction.

SECTION BC 3110
AUTOMATIC VEHICULAR GATES

3110.1 General. Automatic vehicular gates shall comply with the requirements of Sections 3110.2 through 3110.4 and other applicable sections of this code.

‡‡‡ **3110.2 Definition.** The following term is defined in Chapter 2:

VEHICULAR GATE.

3110.3 Vehicular gates intended for automation. Vehicular gates intended for automation shall be designed, constructed and installed to comply with the class of gate in accordance with the requirements of ASTM F 2200.

3110.3.1 Entrapment protection. Defined classes of gates shall be subject to the entrapment protection provisions per UL 325.

3110.4 Vehicular gate operators. Vehicular gate operators, when provided, shall be listed in accordance with UL 325.

SECTION BC 3111
SIDEWALK CAFÉS

3111.1 General. Sidewalk cafés provided beyond the building line shall comply with the requirements of this section, the *New York City Zoning Resolution*, the Commissioners of the Department of Consumer Affairs and Department of Transportation, and with the projection limitations of Chapter 32 of this code.

3111.2 Enclosures. Enclosed sidewalk cafés shall be constructed of noncombustible material. The walls of such enclosures shall not extend more than 8 feet (2438 mm) above the sidewalk. Light-transmitting plastic glazing complying with Section 2606 shall be permitted as glazing within such walls. Light-transmitting plastic skylight glazing complying with Section 2610 may be installed in the roofs of such enclosures.

3111.3 Awnings. Awnings supported entirely from the building may be placed over unenclosed sidewalk cafés provided they are not less than 8 feet (2438 mm) clear above the sidewalk and within the limits specified by the Commissioner of the Department of Consumer Affairs. Such awnings shall be in compliance with Section 3105 of this code.

3111.4 Obstructions prohibited. No part of any awning, enclosure, fixture, equipment or removable platform of a sidewalk café shall be located:

1. Beneath a fire escape so as to obstruct operation of fire escape drop ladders or counter-balanced stairs;
2. So as to obstruct any exit from a building;
3. So as to obstruct any cellar access hatch or areaway;
4. So as to interfere with any vent or other mechanical ventilation outlet or inlet; or
5. So as to interfere with or obscure any standpipe connections, hydrant or associated signage in any way that would hinder its use by the Fire Department.

Exception: Upon special application, the commissioner may permit an easily removable, prominently designated platform, designed in accordance with Section 3111.5, to cover a cellar entrance or areaway that is not used as a required means of egress.

3111.5 Removable platforms. Removable platforms of sidewalk cafés shall be constructed in accordance with the requirements of this section.

3111.5.1 Continuity. Removable platforms shall be constructed to provide for a continuous unbroken and level floor without openings or cracks so as to prevent any material or liquid from falling through to the area beneath.

3111.5.2 Maintenance. No papers, trash or other materials may be permitted to accumulate in the area beneath the floor of any removable platform.

3111.6 Accessibility. Sidewalk cafés and access thereto shall comply with Chapter 11.

3111.7 Assembly seating. Unless separated from seating inside the building by fire partitions complying with Section 713, the seating for enclosed sidewalk cafés shall be added to that inside the building in order to determine whether a place of assembly certificate of operation is required.

3111.8 Rules. In addition to the requirements specified herein, the commissioner may promulgate such additional rules necessary to secure safety.

SECTION BC 3112
FENCES

3112.1 Permitted heights. Fences are permitted to be erected to a maximum height of 10 feet (3048 mm) above the ground.

Exceptions:

1. In residence districts, as established by the *New York City Zoning Resolution*, fences are permitted to be erected to a maximum height of 4 or 6 feet (1829 mm) above the ground, depending upon its location on the zoning lot.
2. Fences in residence districts used in conjunction with nonresidential buildings and public playgrounds, excluding buildings accessory to dwellings, are permitted to be erected to a maximum height of 15 feet (4572 mm) above the ground.
3. Higher fences may be permitted by the commissioner where required for the enclosure of public playgrounds, school yards, parks, and similar public facilities.

SECTION BC 3113
PHOTOVOLTAIC PANELS AND MODULES

3113.1 General. Photovoltaic panels and modules shall comply with the requirements of this code, the *New York City Electrical Code*, the *Zoning Resolution* and the *New York City Fire Code*.

3113.1.1 Rooftop-mounted photovoltaic panels and modules. Photovoltaic panels and modules installed on a roof or as an integral part of a roof assembly shall comply with the requirements of the *New York City Fire Code* and Chapter 15 of this code.

SECTION BC 3114
LARGE WIND TURBINES

3114.1 General. Large wind turbines shall be designed and constructed in accordance with this section.

3114.2 Definitions. The following terms are defined in Chapter 2:

LARGE WIND TURBINE.

LARGE WIND TURBINE TOWER.

3114.3 Design standards. A large wind turbine shall be designed in accordance with standards adopted by rules of the commissioner. Such standards shall include but need not be limited to standards relating to the design of large wind turbines that are developed by the American Wind Energy Association, the New York State Energy Research and Development Authority, the California Energy Commission, the European Wind Turbine Certification, the British Wind Energy Association, the International Electrotechnical Commission, the National Renewable Energy Laboratory, or the Underwriters Laboratory.

3114.4 Wind speed. A large wind turbine shall be designed to withstand winds of up to and including 130 mph (58.1 m/s) or such higher wind load as may be specified in this code or the design standard for such turbine pursuant to Section 3114.3.

3114.5 Brakes and locks. Where necessary for public safety, the commissioner may require that a large wind turbine shall be equipped with a redundant braking system and a passive lock, including aerodynamic overspeed controls and mechanical brakes.

3114.6 Visual appearance. A large wind turbine shall be white, off-white, grey, or another non-obtrusive color specified by the commissioner.

‡‡‡ **3114.7 Access.** Access to a large wind turbine shall be limited as follows:

1. Access to electrical components of a large wind turbine shall be prevented by a lock.
2. A large wind turbine tower shall not be climbable, except by authorized personnel, up to a height of 10 feet (3048 mm) measured from the base of such tower.

‡‡‡ **3114.8 Noise.** A large wind turbine shall be designed to comply with the sound level limit of Section 24-232.1 of the *Administrative Code*.

‡‡‡ **3114.9 Shadow flicker.** The commissioner shall by rule establish shadow flicker limitations for large wind turbines for the purpose of limiting, to the extent practicable, such flicker on buildings adjacent to such turbines.

‡‡‡ **3114.10 Signal interference.** The commissioner shall establish rules governing large wind turbines for purpose of minimizing, to the extent practicable, interference by such turbines with radio, telephone, television, cellular or other similar signals.

‡‡‡ **3114.11 Setback.** No part of a large wind turbine or large wind turbine tower shall be located within a horizontal distance of a property line that is equal or less than one-half the height of such turbine, including such tower, measured from the base of such tower or, if there is no such tower, the base of such turbine.

Exception: A turbine or tower for which each owner of property adjacent to such property line has entered into a written agreement providing that such turbine or tower or a part thereof may be located closer to such property line than this section allows.

SECTION BC 3115
SMALL WIND TURBINES

3115.1 General. In addition to other applicable requirements in this code, other law or rule, and established by the commissioner, small wind turbines shall be designed and constructed in accordance with this section.

3115.2 Definitions. The following terms are defined in Chapter 2:

SMALL WIND TURBINE.

SMALL WIND TURBINE TOWER.

3115.3 Design standards. A small wind turbine shall be designed in accordance with standards adopted by rules of the commissioner. Such standards shall include but need not be limited to standards relating to the design of small wind turbines that are developed by the American Wind Energy Association, the New York State Energy Research and Development Authority, the California Energy Commission, the Small Wind Certification Council, the British

Wind Energy Association, the International Electrotechnical Commission, the National Renewable Energy Laboratory, or the Underwriters Laboratory.

3115.4 Wind speed. A small wind turbine shall be designed to withstand winds of up to and including 130 mph (58.1 m/s) or such higher wind load as may be specified in this code or the design standard for such turbine pursuant to Section 3113.3.

3115.5 Brakes and locks. Where deemed necessary by the commissioner, a small wind turbine shall be equipped with a redundant braking system and a passive lock, including aerodynamic overspeed controls and mechanical brakes.

3115.5.1 Locking before hurricane or strong wind conditions. If a hurricane or strong wind conditions are expected, the commissioner may order that small turbines equipped with passive locks be stopped and locked.

3115.6 Visual appearance. A small wind turbine shall be white, off-white, grey, or another non-obtrusive color specified by the commissioner.

3115.7 Lighting. A small wind turbine shall not be artificially lighted.

Exception: Lighting that is required by this code or other applicable laws or rules, provided that such lighting is shielded in accordance with rules promulgated by the commissioner.

3115.8 Access. Access to a small wind turbine shall be limited as follows:

1. Access to electrical components of a small wind turbine shall be prevented by a lock.
2. A small wind turbine tower shall not be climbable, except by authorized personnel, up to a height of 10 feet (3048 mm) measured from the base of such tower.

3115.9 Noise. A small wind turbine shall be designed in accordance with applicable sections of the *New York City Noise Control Code*.

3115.10 Shadow flicker. The commissioner shall by rule establish shadow flicker limitations for small wind turbines for the purpose of limiting, to the extent practicable, such flicker on buildings adjacent to such turbines.

3115.11 Signal interference. The commissioner shall establish rules governing small wind turbines for purpose of minimizing, to the extent practicable, interference by such turbines with radio, telephone, television, cellular or other similar signals.

3115.12 Setback. No part of a small wind turbine or small wind turbine tower shall be located within a horizontal distance of a property line that is equal or less than one-half the height of such turbine, including such tower, measured from the base of such tower or, if there is no such tower, the base of such turbine.

Exception: Turbines or towers for which each owner of property adjacent to such property line has entered into a written agreement providing that such turbine or tower or a part thereof may be located closer to such property line than this section allows.